

Joel Thomas

Full Stack Developer - Python, React, AI

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GitHub: github.com/joelt006 | Portfolio: joelt006.github.io/Portfolio-site

RAG project: github.com/joelt006/rag-qa-assistant

PROFESSIONAL SUMMARY

Informatics graduate (bachelor thesis defended, graded 85%) and full-stack developer who builds and ships complete applications end to end. Built a full digital banking platform with a Django REST backend, a Vue 3 frontend, and an on-device machine-learning assistant, plus a published retrieval-augmented generation (RAG) AI application. Skilled in Python, Vue and React, REST API design, SQL, Git, Docker and cloud deployment, with a strong focus on security and clean, maintainable code. Full labour-market access in Slovakia under Section 23a (no work permit required). Available immediately.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, HTML, CSS (learning TypeScript)

Frontend: Vue 3 (Vuex, Vue Router), React, responsive design, single-page applications, REST and JSON integration

Backend: Django, Django REST Framework, Flask, REST API design, JWT and OTP authentication, object-oriented design

AI and Machine Learning: LLM applications, retrieval-augmented generation (RAG), embeddings, vector databases, LangChain; scikit-learn (TF-IDF, Logistic Regression) for local intent classification

Databases: PostgreSQL, SQLite, SQL, relational schema design, atomic transactions

DevOps and Cloud: Docker, Microsoft Azure, Git and GitHub, Linux, build and deploy automation, CI/CD principles

Security: JWT auth, OTP, PBKDF2/HMAC hashing, rate limiting, session management, authorization design

Practices: Automated testing, clean code, technical documentation, Agile collaboration, AI-assisted development. English: C1 (Advanced)

PROJECTS

MyBank - Full-Stack Banking Platform with Local AI Assistant

Django REST Framework, Vue 3, SQL, JWT/OTP, scikit-learn

- Built a complete digital banking platform end to end: accounts, money transfers, beneficiaries, scheduled transfers, loans (with live EMI and repayment tracking), fixed deposits, card management, notifications, customer support, and a full admin portal.
- Built Heva, an on-device machine-learning assistant (scikit-learn TF-IDF + Logistic Regression) that classifies banking requests across 17 intents at 93.4% cross-validated accuracy - runs locally with no LLM and no external API, so no customer data leaves the system.
- Engineered strong security: JWT auth with email OTP, PBKDF2-hashed transaction PINs, CVVs derived on demand via HMAC (never stored), OCR/KYC identity checks, rate limiting, session management, and a read-only, confirmation-gated authorization model for the assistant.
- Designed the REST/JSON API and data model (15+ entities), enforced atomic transfers with row-level locking, and documented endpoints, models, and the assistant security contract.

Document Q&A Assistant - AI Application (RAG)

Python, LangChain, Chroma vector database, OpenAI, Ollama, Streamlit

- Built and published an AI application that answers natural-language questions grounded in a document's content, showing source passages to prevent hallucination; implemented the full RAG pipeline (chunking, embeddings, vector store, retrieval, LLM answering) with a backend switchable between a cloud and a local model.

CloudNotes - Containerized Cloud Application

Python, Flask, PostgreSQL, Docker, Microsoft Azure

- Deployed a multi-container Flask and PostgreSQL application to Microsoft Azure using Docker and automated provisioning scripts.

EDUCATION

Bachelor of Science in Informatics - Technical University of Kosice, Slovakia (2026)

- Thesis defended, graded 85%. Shipped a feature into an active research codebase (Three.js, WebXR), with full developer documentation.